

DATA STRUCTURE

LAB PROGRAM

22. Write a C program to Graph traversal using Depth First Search

```
#include <stdio.h>
```

```
#define MAX 20
```

```
int graph[MAX][MAX];
```

```
int visited[MAX];
```

```
int n;
```

```
void DFS(int vertex)
```

```
{
```

```
    int i;
```

```
    visited[vertex] = 1;
```

```
    printf("%d ", vertex);
```

```
    for(i = 0; i < n; i++)
```

```
    {
```

```
        if(graph[vertex][i] == 1 && !visited[i])
```

```
        {
```

```
            DFS(i);
```

```
    }  
  }  
}
```

```
int main()
```

```
{
```

```
    int i, j, start;
```

```
    printf("Enter the number of vertices: ");
```

```
    scanf("%d", &n);
```

```
    printf("Enter the adjacency matrix:\n");
```

```
    for(i = 0; i < n; i++)
```

```
    {
```

```
        for(j = 0; j < n; j++)
```

```
        {
```

```
            scanf("%d", &graph[i][j]);
```

```
        }
```

```
    }
```

```
    for(i = 0; i < n; i++)
```

```
        visited[i] = 0;
```

```
printf("Enter the starting vertex (0 to %d): ", n - 1);  
scanf("%d", &start);  
  
printf("DFS Traversal: ");  
DFS(start);  
  
printf("\n");  
  
return 0;  
}
```

Output

```
Enter the number of vertices: 2  
Enter the adjacency matrix:  
1 2  
1 9  
Enter the starting vertex (0 to 1): 1  
DFS Traversal: 1 0
```

=== Code Execution Successful ===